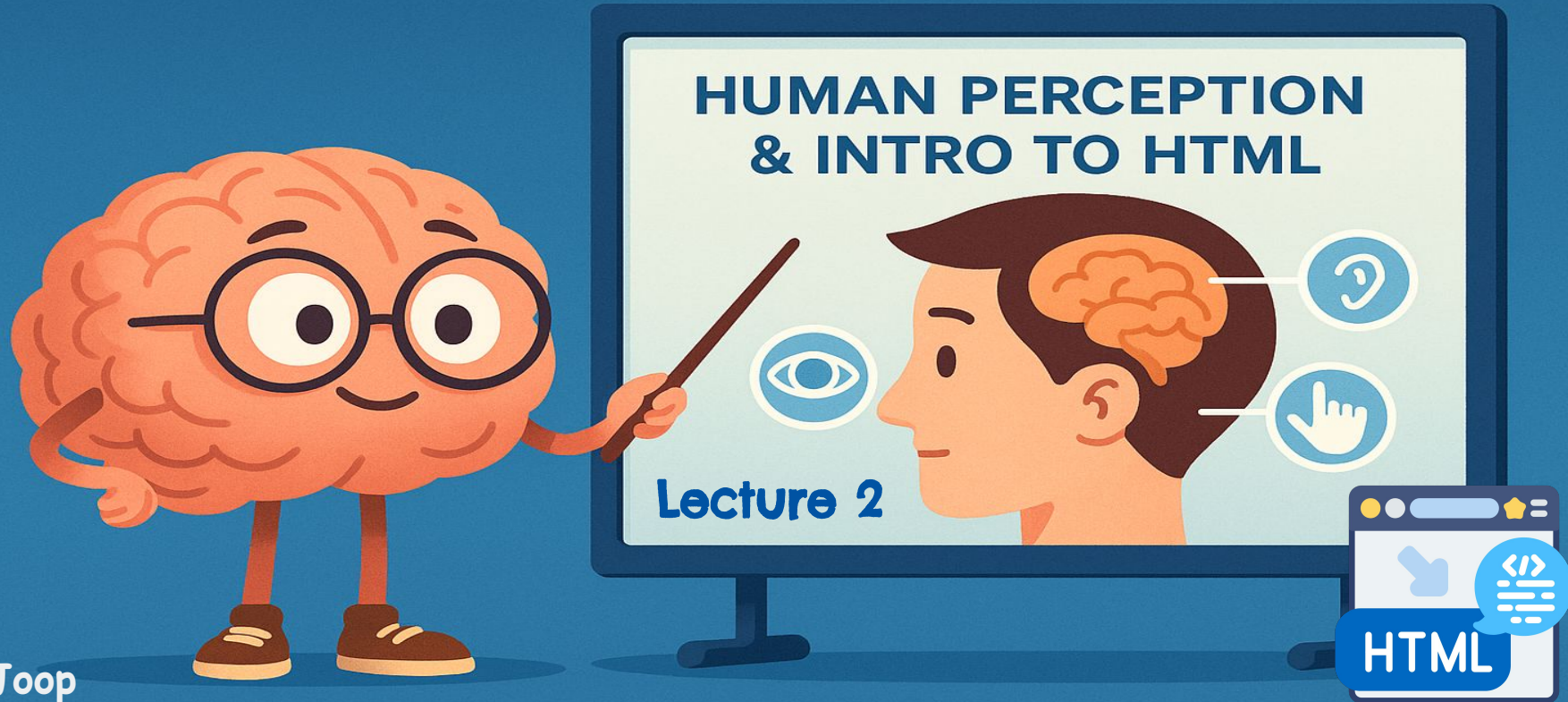


Web User Experience & Database Design



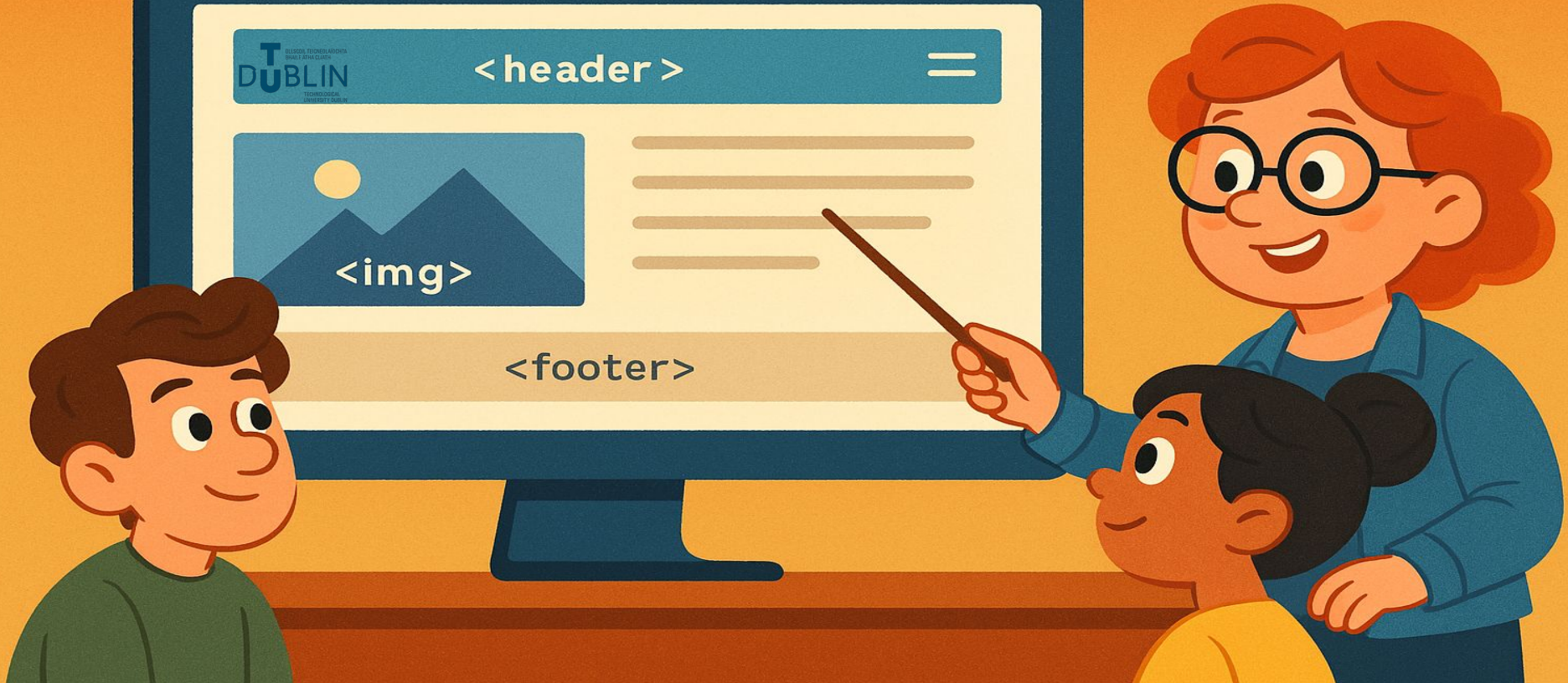
By Joop
Inthraporn Aranyanak, PhD

Lecture 2 - Lesson Overview

- Introduction to HTML
-
- What is Human Perception?
- Visual Perception
- Hearing Perception
- Touch Perception
- Taste Perception
- Olfactory Perception



INTRODUCTION TO HTML



Web Standards and the W3C Consortium

- W3C – World Wide Web Consortium
 - Develops recommendations and prototype technologies related to the Web
 - Produces specifications, called Recommendations, in an effort to standardise web technologies
- WAI – Web Accessibility Initiative
 - This is a part of W3C that focuses on making sure the web is usable for everyone—including people with disabilities.





Free Resources to Learn HTML & CSS

-  1. **Code.org's Web Lab:** <https://code.org/en-US/tools/web-lab>
-  2. **W3Schools:** <https://www.w3schools.com/>
-  3. **HTML Dog:** <https://htmldog.com>
-  4. **freeCodeCamp:** <https://www.freecodecamp.org/learn>
-  5. **CodePen (Playground):** <https://codepen.io>



WEBDH1002: Web and UI Design

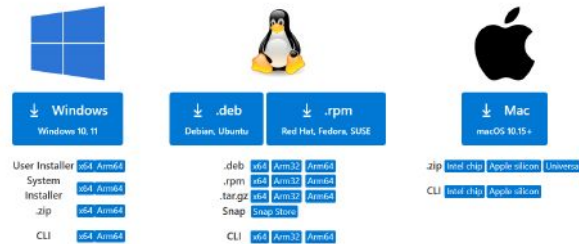
Install Visual Studio Code (VS Code)

Instructions:

1. Download VS code here: <https://code.visualstudio.com/Download>

Download Visual Studio Code

Free and built on open source. Integrated Git, debugging and extensions.



You must read the document on how to install the **Live Server** and **Prettier** extensions before the lab class.

What is HTML?

HyperText Markup Language

- The 'Language' of the World Wide Web
- Hypertext is text which links to other text or links to another page

Example: www.tudublin.ie

- Markup is the code ('tags') used to give meaning to page content (text or images or video) and give it structures, e.g.
 - Define a page header and paragraph text.
 - Show financial data in a table.
- Basically, markup tells the browser what each part of the content is.



HTML Tags

- Tags are used to assign meaning to page content
- They are keywords described by the HTML standard and are denoted by angle brackets
 - `<p>The tags make this a paragraph</p>`
 - `<h1>This is a level 1 header</h1>`
- Most HTML tags, like the ones above, open and close
- Others such as `<br>` (a line break), known as void elements, are self contained and not closed
- Tags may contain attributes which are used to provide more information or content
 - The `<img>` tag has a `src` attribute which refers to the image to be displayed and `alt` for alternate text
 - ``



HTML Tags

- HTML tags are keywords surrounded by angle brackets like `<html>`
- HTML tags normally come in pairs like `<p>` and `</p>`

HTML Tags structure

`<element> ..... </element>`



Start Tag



Content



End Tag



HTML Tags

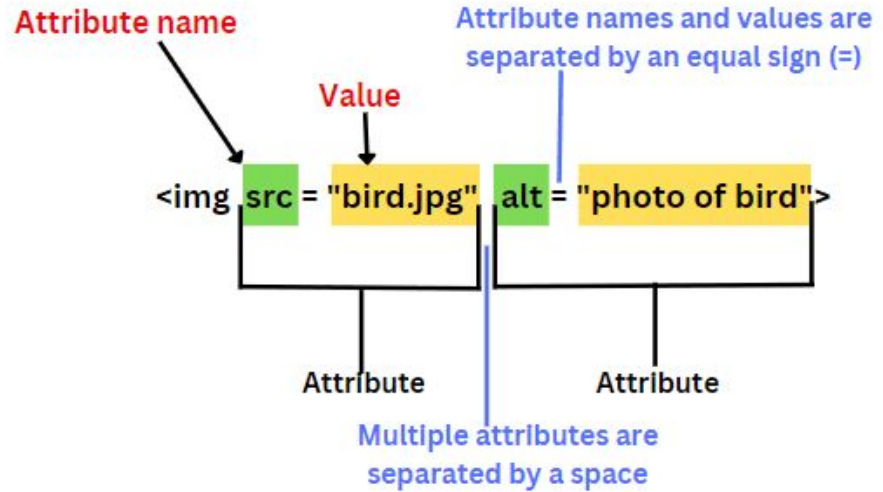
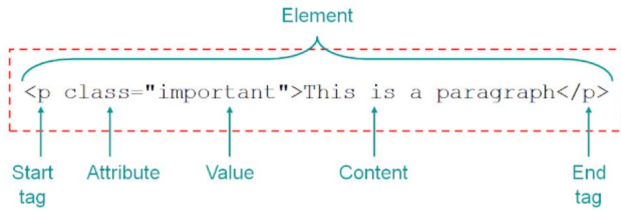


Fig: An img element with two attributes

Setting Up Your Project Folder

When you start a new project, it's best to create a main folder – for example, **on your desktop** – and keep all your files inside it. Put your HTML, CSS, JavaScript, and images together in this folder.

Why We Use **“index.html”** as the Main Page

When you create a new web project, you usually name the main page **index.html** because most web servers automatically look for a file with that name when someone visits your site.

HTML Page Structure

- The current standard is HTML5
- Here is a template on which all your HTML5 pages should be based on
 - i.e. all new pages should contain the following sections and elements
- When you open VS Code to start writing HTML, there's a quick shortcut to save time. Just type "!" and press Enter (or Tab).
- VS Code will automatically create the basic HTML structure for you – including the `<html>`, `<head>`, and `<body>` tags.
- Most of the time, you write your main content inside the `<body>` tag.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8"> <!-- Character encoding -->
  <title>Page Title</title> <!-- Title shown on browser tab -->
</head>
<body>
  <!-- Main content of the page goes here -->
</body>
</html>
```

HTML Page Structure

<!doctype html>

Tells the browser that this page contains HTML5

<html lang="en">

The root of the page; lang="en" means the content is in English

<head>

Contains page information (metadata) like title, styles, link an external CSS file, and some JavaScript files

<meta charset="UTF-8">

Defines character encoding (UTF-8 supports most characters and symbols)

<title>

Sets the title of the page (appears on the browser tab)

<body>

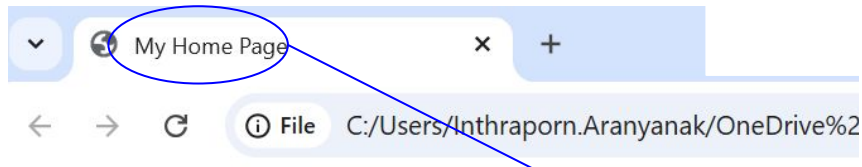
Contains everything that is displayed on the page (text, images, links, etc.)

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8"> <!-- Character encoding -->
  <title>Page Title</title> <!-- Title shown on browser tab -->
</head>
<body>
  <!-- Main content of the page goes here -->
</body>
</html>
```

How can I write my first web page?

1. Open a text editor e.g., Notepad++, TextPad
2. Put in code from the Example HTML5 Web Page (the previous) Slide
3. Add `<h1>Welcome to My Website</h1>` and `<p>This is a basic HTML structure.</p>` between the `<body></body>` tags.
4. Save it as **index.html** on your Desktop
5. Double click on the **index.html** where it is located
6. Your first HTML page will open in your web browser!

HTML Basic Structure



Welcome to my website!

This is a basic HTML structure using Notepad

```
index.html
File Edit View

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>My Home Page</title>
</head>
<body>
  <h1>Welcome to my website!</h1>
  <p>This is a basic HTML structure using Notepad</p>
</body>
</html>
```



Try it yourself

HTML - Some Guidelines

- Usually HTML pages are accessed from a web server – however pages can also be opened ‘locally’ in a browser
- By convention, the main or **home page should be called index.html** - Most web servers (like Apache, Nginx, IIS) automatically load index.html as the default page.
- Avoid spaces in HTML file names as they can be problematic when accessing from a web server
- When editing, always close tags immediately
 - E.g. follow `<p>` with `</p>` then fill in the gaps
- Ensure that Windows is not set to ‘hide file extensions for known files’

HTML - Some Guidelines

HTML is not case-sensitive. However, it is recommended to use lowercase for HTML tags and attributes

It is important to remember that most web servers are case sensitive with file names

- **Index.html is not the same as index.html**
- It's a good idea to adopt a naming convention:
 - As a general rule use lower case for names and extensions
 - **Avoid spaces**
 - Always follow the same naming convention, for example:
 - Keep file names lower cased e.g. index.html
 - If there is more than one word in the file name, separate words by – e.g. contact-us.html
 - An alternative is to use **underscores_instead_of_spaces.html**

The Heading Element

`<h1>Heading Level 1</h1>`

`<h2>Heading Level 2</h2>`

`<h3>Heading Level 3</h3>`

`<h4>Heading Level 4</h4>`

`<h5>Heading Level 5</h5>`

`<h6>Heading Level 6</h6>`

Heading Level 1

Heading Level 2

Heading Level 3

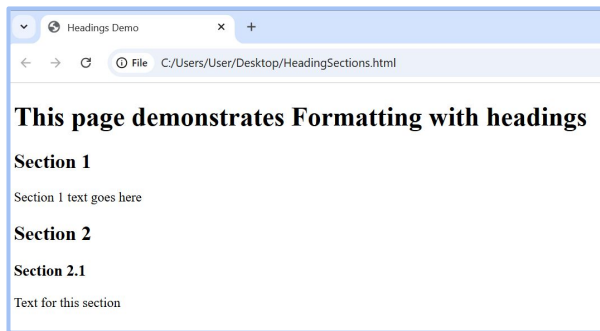
Heading Level 4

Heading Level 5

Heading Level 6

Formatting: Headings

```
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <title>Headings Demo</title>
  </head>
  <body>
    <h1>This page demonstrates Formatting with headings</h1>
    <h2>Section 1</h2>
      <p>Section 1 text goes here</p>
    <h2>Section 2</h2>
      <h3>Section 2.1</h3>
        <p>Text for this section</p>
      </h3>
    </h2>
  </body>
</html>
```



Headings are used to divide the page
Into sections

There are 6 levels **<h1>...** **<h6>**

Use as you would headings in
Microsoft Word

<h1> should only be used once

Paragraph & Line Break Elements

Paragraph element: `<p> ...paragraph goes here... </p>`

- Groups sentences and sections of text together.
- Block Display – Configures empty space above and below

Line Break element: `...text goes here <br>`

- Stand-alone, or void tag
- Causes the next element or text to display on a new line
- For example:

`<p>This is the first line.<br>This is the second line.</p>`

This is the first line.
This is the second line.

HTML Lists

- Unordered List: () – for bullet points
- Ordered List: () – for numbered items
- Description List: (<dl>) – for terms and their descriptions

An unordered list:

- This is a list item
- This is another list item
- This is one more list item

An ordered list:

1. This is List item number 1
2. This is List item number 2
3. This is List item number 3

Description Lists:

- | | |
|-------|--------------|
| Tea | - hot drink |
| Juice | - cold drink |

HTML- Unordered List

- `<ul>` Displays a bullet, or list marker, before each entry in the list. `</ul>`
- The items in unordered lists are marked with bullets (small black circles) by default.
 - `<li>` defines each item in a list `</li>`

However, the default bullet style for the list items can be changed using a type attribute.

Such as, `<ul type="square">`, `<ul type="circle">`

```
<body>
  <ul>
    <li>List item </li>
    <li>List item</li>
    <li>List item</li>
  </ul>
```

- List item
- List item
- List item

```
<body>
  <ul type="circle">
    <li>List item </li>
    <li>List item</li>
    <li>List item</li>
  </ul>
```

- List item
- List item
- List item

```
<ul type="square">
  <li>List item</li>
  <li>List item</li>
  <li>List item</li>
</ul>
</body>
```

- List item
- List item
- List item

HTML- Ordered List

- `<ol>` defines an ordered (numbered) list`</ol>`
- Displays a numbering or lettering system to itemise the information contained in the list, default is numerals.
 - `<li>` defines each item in a list `</li>`

If you want to create ordered list with alphabet or Roman numbers, you just need to add `type="a"` or `type="I"` to the `<ol>` tag

```
<body>
  <h1>An ordered list:</h1>
  <ol>
    <li>This is List item number 1</li>
    <li>This is List item number 2</li>
    <li>This is List item number 3</li>
  </ol>
</body>
```

An ordered list:

1. This is List item number 1
2. This is List item number 2
3. This is List item number 3

```
<body>
  <h3>An alphabetized list (lowercase letters):</h3>
  <ol type="a">
    <li>Peach</li>
    <li>Apricot</li>
  </ol>
  <h3>A numbered list (Roman numerals):</h3>
  <ol type="I">
    <li>Peach</li>
    <li>Apricot</li>
  </ol>
</body>
```

An alphabetized list (lowercase letters):

- a. Peach
- b. Apricot

A numbered list (Roman numerals):

- I. Peach
- II. Apricot

HTML- Description List

Useful to display a list of terms and descriptions or a list of FAQ and answers. To create a description list, we use the `<dl>` tag. This tag comes in pairs.

- `<dl>` defines a definition list
- `<dt>` defines the term
- `<dd>` defines detail of term

```
<body>
  <h1>Description Lists:</h1>
  <dl>
    <dt>Tea</dt>
    <dd>- hot drink</dd>
    <dt>Juice</dt>
    <dd>- cold drink</dd>
  </dl>
</body>
```

Description Lists:

Tea
- hot drink

Juice
- cold drink

Logical Style Elements

Define the semantic or structural meaning of the text they enclose. These tags emphasise the intended meaning of the content rather than just defining how it looks. They are useful for accessibility and for search engines to better understand your content.

- `<strong>`, usually displayed as **bold**.
- `<em>`, usually displayed in *italic*.
- `<q>`, usually displayed with “quotation marks”.
- `<abbr>` defines an abbreviation or acronym.

```
1 <!DOCTYPE html>
2 <html>
3   <head>
4     <title>Title of the document</title>
5   </head>
6   <body>
7     <p>This is <strong>very important</strong> information.</p>
8     <p>This is <em>emphasized</em> text.</p>
9     <p>She said, <q>This is amazing!</q></p>
10    <p>The <abbr title="World Health Organization">WHO</abbr> is a global he:
11  </body>
12 </html>
```

This is **very important** information.

This is *emphasized* text.

She said, “This is amazing!”

The WHO is a global health organization.

HTML Links

- HTML links connect web pages, sections, files, or external resources.
- HTML links, created using the `<a>` tag (called an anchor element)
- Hyperlinks are applied to a phrase, a word, an image or any HTML element.
- Syntax: `<a href="URL">Link Text</a>`

Link example

[W3Docs.com](https://www.w3docs.com/)
[Visit Google](https://www.google.com/)

```
<!DOCTYPE html>
<html>
  <head>
    <title>Title of the document</title>
  </head>
  <body>
    <h2>Link example</h2>
    <a href="https://www.w3docs.com/">W3Docs.com</a> <br>
    <a href="https://www.google.com" target="_blank" title="Search on Google">Visit Google</a>
  </body>
</html>
```

Diagram illustrating the HTML link syntax with annotations:

- Link** (points to the `href` attribute)
- Your text** (points to the link text `W3Docs.com`)
- New line** (points to the `<br>` tag)
- displayed as a tooltip on hover** (points to the `title` attribute)
- open the link in a new tab** (points to the `target="_blank"` attribute)

Link example

[W3Docs.com](https://www.w3docs.com/)
[Visit Google](https://www.google.com/)

Search on Google

HTML - Adding Image & Break Line

- HTML images are used to display pictures on a webpage.
- Images are inserted using the tag (called an image element).
- Images can be from local files or external URLs.
- The alt attribute provides alternative text for accessibility and when the image cannot be displayed.
- **Syntax:**

```
<!DOCTYPE html>
<html>
  <head>
    <title>Title of the document</title>
  </head>
  <body>
    <h2>This is my favorite place</h2>
    <img src = "Beach.jpg" alt = "A picture of a beach">
    <p>I really enjoy the beach when I am in Thailand.<br> This is my favorite
      place in Phuket.</p>
  </body>
</html>
```

This is my favorite place



I really enjoy the beach when I am in Thailand.
This is my favorite place in Phuket.

Div Element

The `<div>` element in HTML is a block-level container used to group and organise content. It has no specific semantic meaning but serves as a generic container for styling or layout purposes when combined with CSS or JavaScript.

- Syntax: **`<div>Content goes here</div>`**
- Key Features:
 - Generic Container: Used to group other elements (text, images, forms, etc.).
 - Styling and Layout: **Commonly styled with CSS for layout or design purposes.**
 - Block-Level Element: Takes up the full width of its parent container by default.

```
1 <!DOCTYPE html>
2 <html>
3   <head>
4     <title>The <div> Tag</title>
5   </head>
6   <body>
7     <h1> The <div> Tag </h1>
8     <div style="background-color: lightblue; padding: 10px;">
9       <p>This Section 1 has a blue background color and 10px padding.</p>
10    </div>
11    <br>
12    <div style="background-color: red; padding: 10px;">
13      <p>This Section 2 has a red background color and 10px padding.</p>
14    </div>
15  </body>
16 </html>
```

The `<div>` Tag

This Section 1 has a background color in blue and padding.

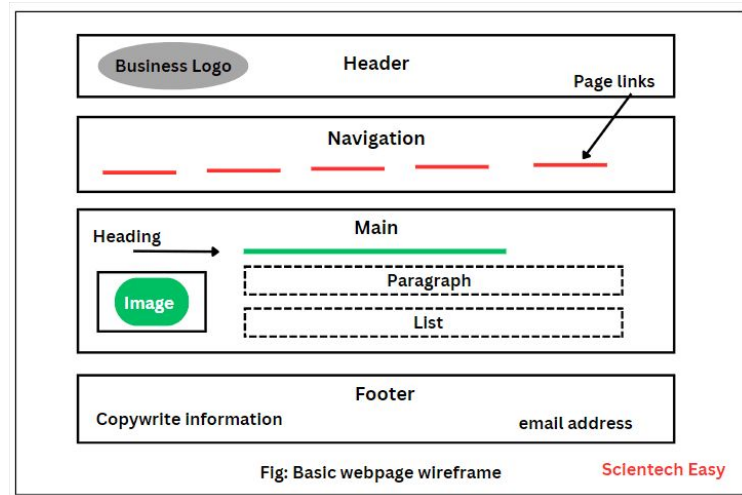
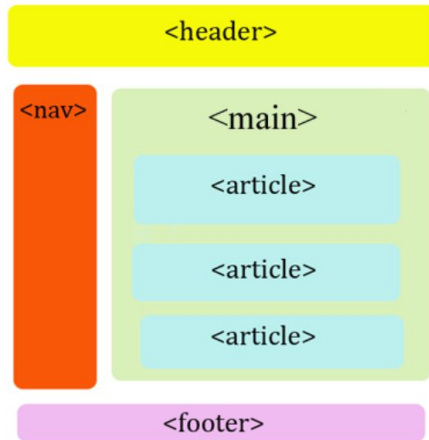
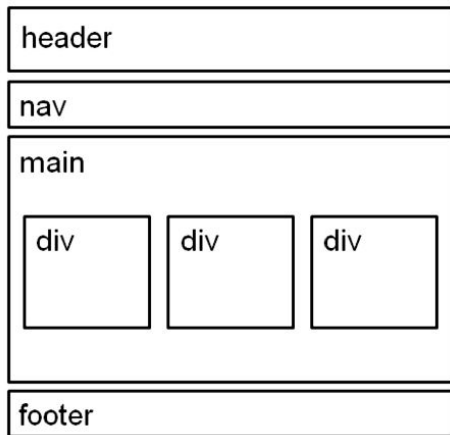
This Section 2 has a background color in red and padding.

The `<div>` Tag with Flexbox

This Section 1 has a blue background color and 10px padding.

This Section 2 has a red background color and 10px padding.

HTML5 Structural Elements



HTML5 structural elements are semantic tags that organise content, improve accessibility, and enhance search engine understanding.

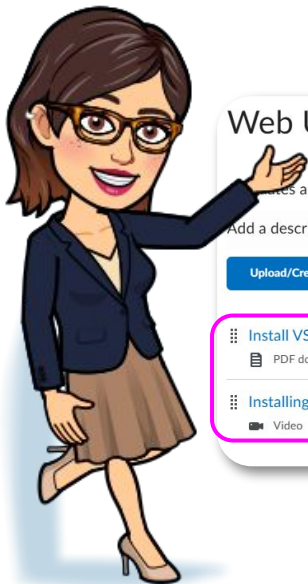
```

<body>
<header>document headings go here </header>
<nav>main navigation goes here</nav>
<main>main content goes here</main>
<footer>document footer information goes
here</footer>
</body>
  
```

For the following examples, I've already uploaded the code to Brightspace.

Watch the Installing VS Code tutorial to get started.

Please try the tutorial code at home before the lab session!



Web UX Lab 2
Print
Settings

Add dates and restrictions...

Add a description...

Upload/Create Existing Activities Bulk Edit

Install VS Code
PDF document

Installing VS Code
Video

Web UX Lecture 2
Print
Settings

Add dates and restrictions...

Add a description...

Upload/Create Existing Activities Bulk Edit

TutorialCode_Week2(Web UX)-20260915T164012Z-1-001
Zip Compressed File



My First Web Page

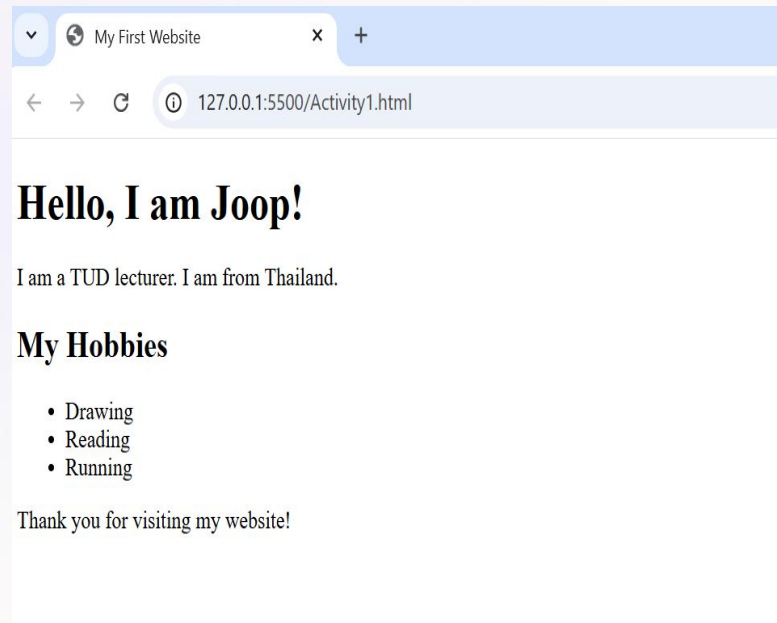
```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width,initial-scale=1.0">
  <title>My First Website</title>
</head>
<body>

  <h1>Hello, I am Joop!</h1>
  <p>I am a TUD lecturer. I am from Thailand.</p>

  <h2>My Hobbies</h2>
  <ul>
    <li> Drawing </li>
    <li> Reading </li>
    <li> Running </li>
  </ul>

  <p>Thank you for visiting my website!</p>

</body>
</html>
```





Adding a New Section & Link

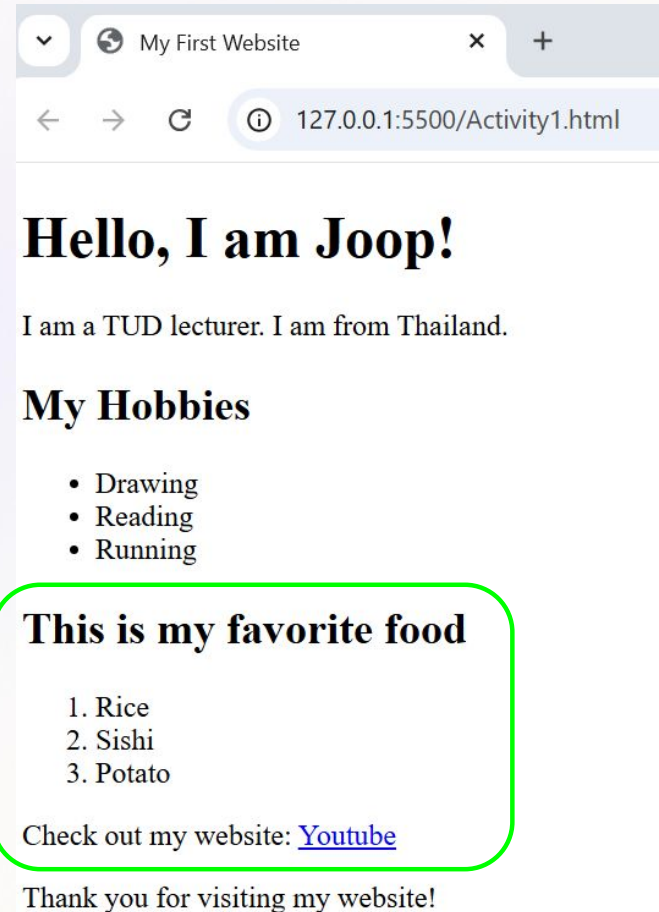
```
<body>
  <h1>Hello, I am Joop!</h1>
  <p>I am a TUD lecturer. I am from Thailand.</p>

  <h2>My Hobbies</h2>
  <ul>
    <li> Drawing </li>
    <li> Reading </li>
    <li> Running </li>
  </ul>

  <h2>This is my favorite food</h2>
  <ol>
    <li>Rice</li>
    <li>Sishi</li>
    <li>Potato</li>
  </ol>

  <p>Check out my website: <a href = "https://www.youtube.com/"
target = "_blank">Youtube</a></p>

  <p>Thank you for visiting my website!</p>
</body>
```





Adding Image & Break Line

```
<h2>This is my favorite food</h2>
```

```
<ol>
```

```
<li>Rice</li>
```

```
<li>Sishi</li>
```

```
<li>Potato</li>
```

```
</ol>
```

```
<p>Check out my website: <a href = "https://www.youtube.com/"  
target = "_blank">Youtube</a></p>
```

```
<h2>This is my favorite place</h2>
```

```
<img src = "Beach.jpg" alt = "A picture of a beach">
```

```
<p>I really enjoy the beach when I am in Thailand.<br> This is my  
favorite place in Phuket.</p>
```

```
<p>Thank you for visiting my website!</p>
```





Adding a Video

```
<h2>This is my favorite place</h2>
```

```
<img src = "Beach.jpg" alt = "A picture of a beach">
```

```
<p>I really enjoy the beach when I am in Thailand.<br> This is my  
favorite place in Phuket.</p>
```

```
<h2>Sample Video Clip</h2>
```

```
<video width="400" controls>
```

```
<source src="Clip.mp4" type="video/mp4">
```

Your browser does not support the video tag.

```
</video>
```

```
<p>Thank you for visiting my website!</p>
```

This is my favorite food

1. Rice
2. Sishi
3. Potato

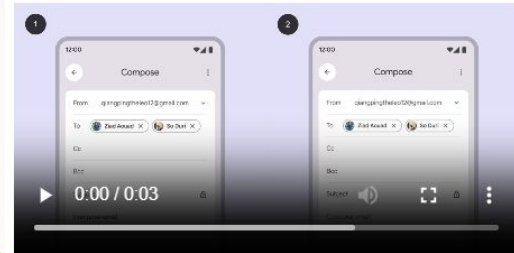
Check out my website: [Youtube](#)

This is my favorite place



I really enjoy the beach when I am in Thailand.
This is my favorite place in Phuket.

Sample Video Clip



Thank you for visiting my website!



Checking your attendance

Week 2 Lecture 1

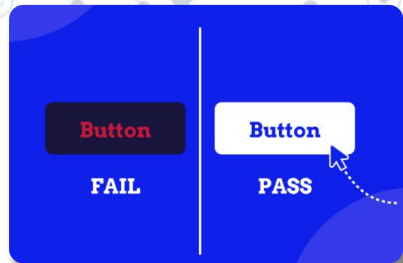


Note that you will get 1% for answering the question correctly.

End of Part 1



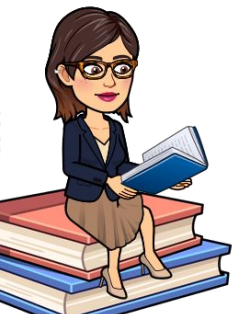
WEB UX DESIGN



Which button should we press?

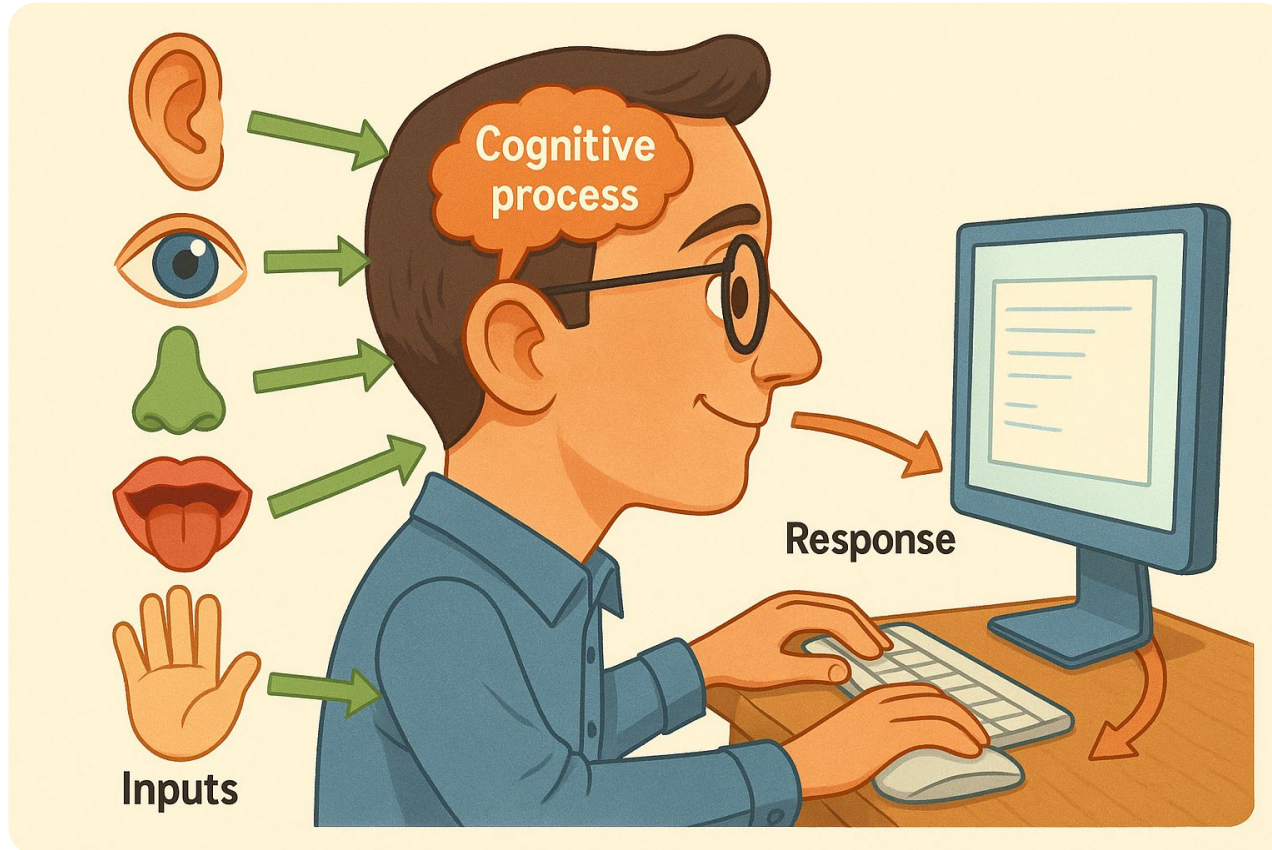
“

Have you ever gotten lost while using a website or app?

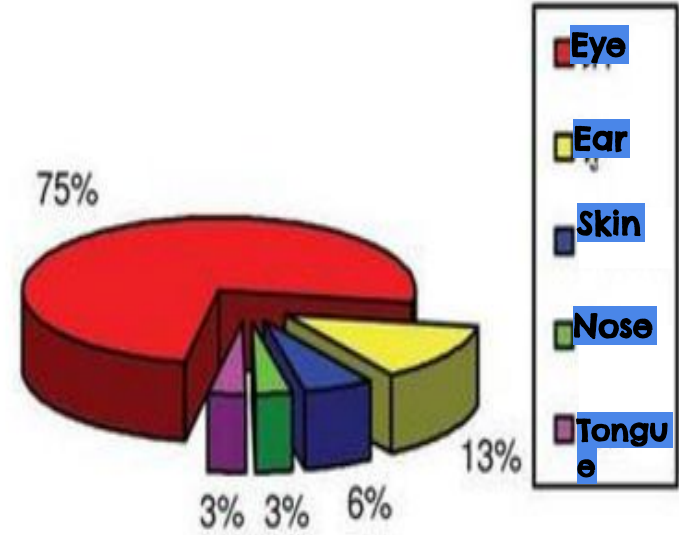
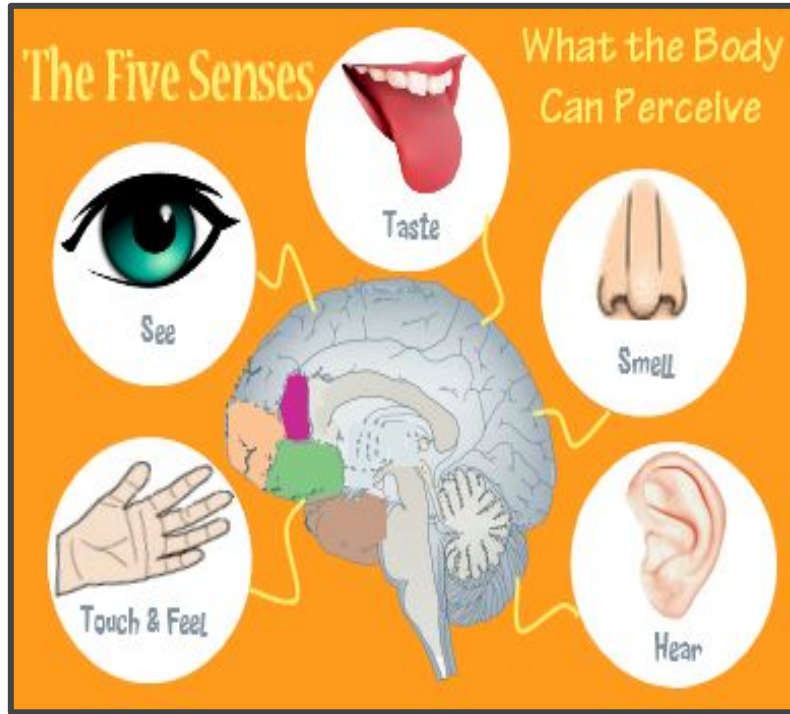


<https://uxmovement.com/buttons/the-myths-of-color-contrast-accessibility/>
<https://ie.pinterest.com/akajonep/bad-elevator-buttons/>
<https://www.uxmatters.com/mt/archives/2017/03/3-kinds-of-simplicity.php>

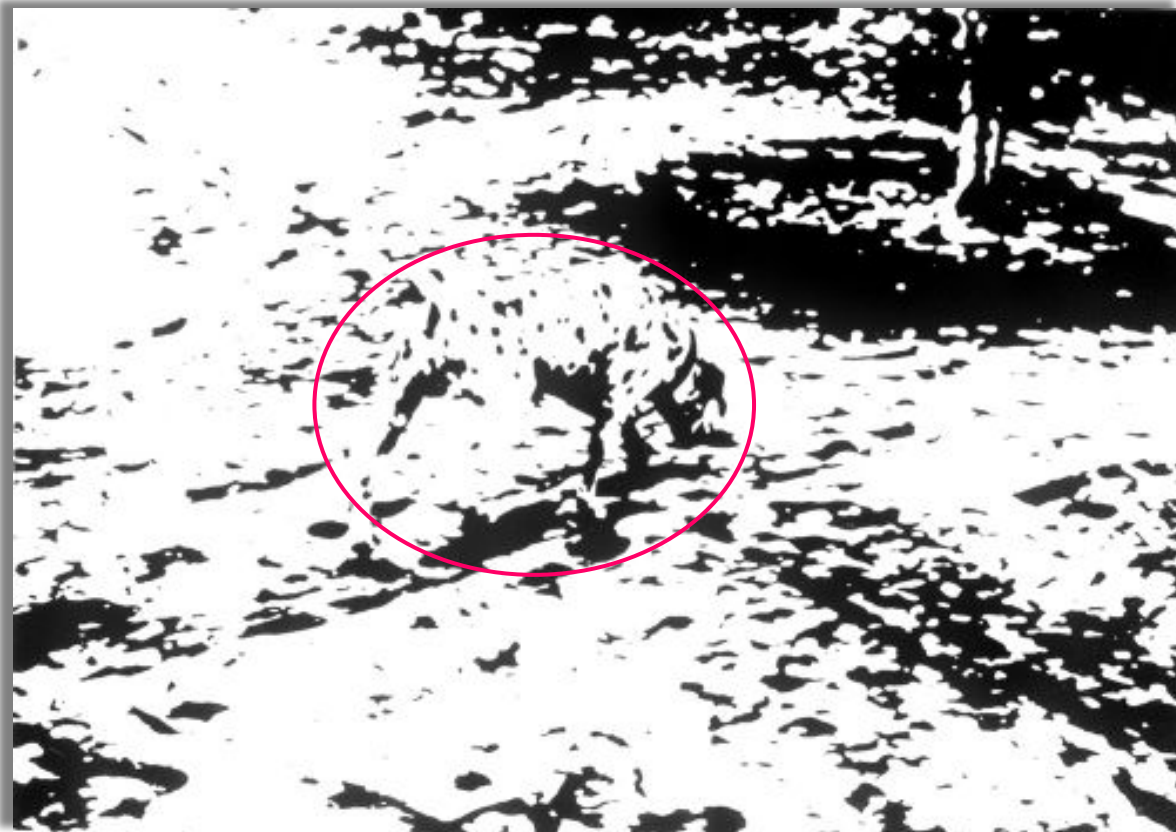
In the process of human perception



Human Perception



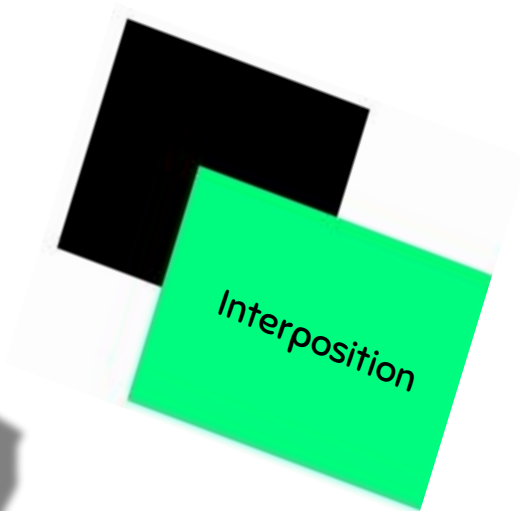
Do you see any animals in the picture?



[More Here>>](#)

These factors help human perceive depth

Texture

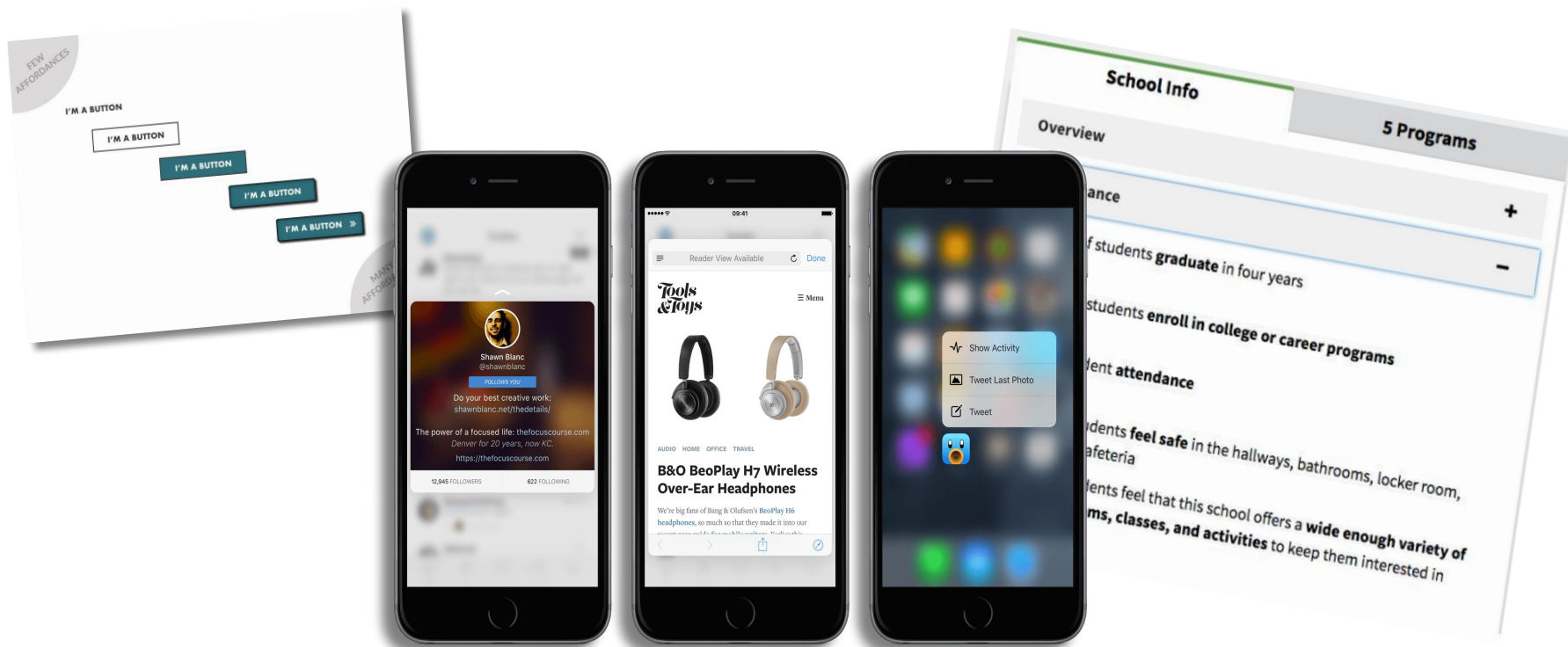


Perceptive

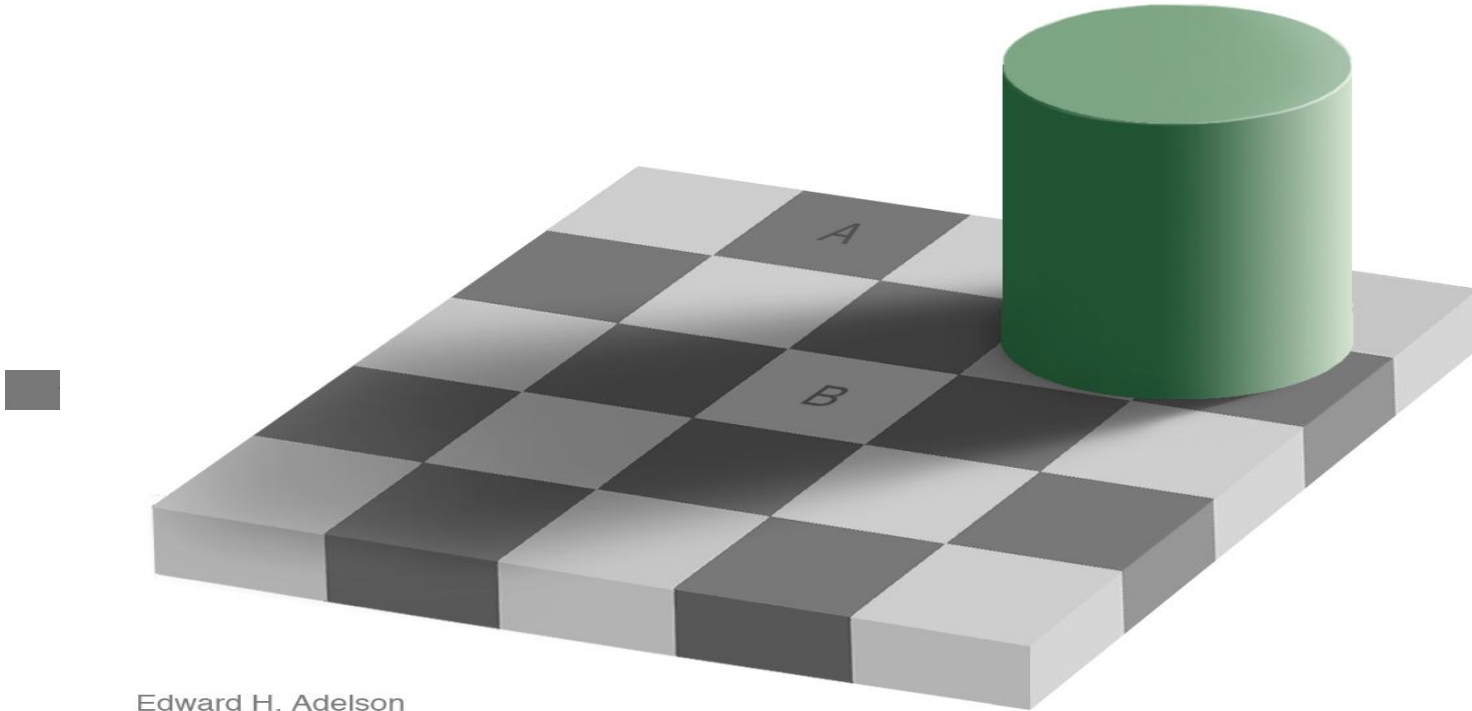
These elements all contribute to how we perceive depth



Depth in interface design



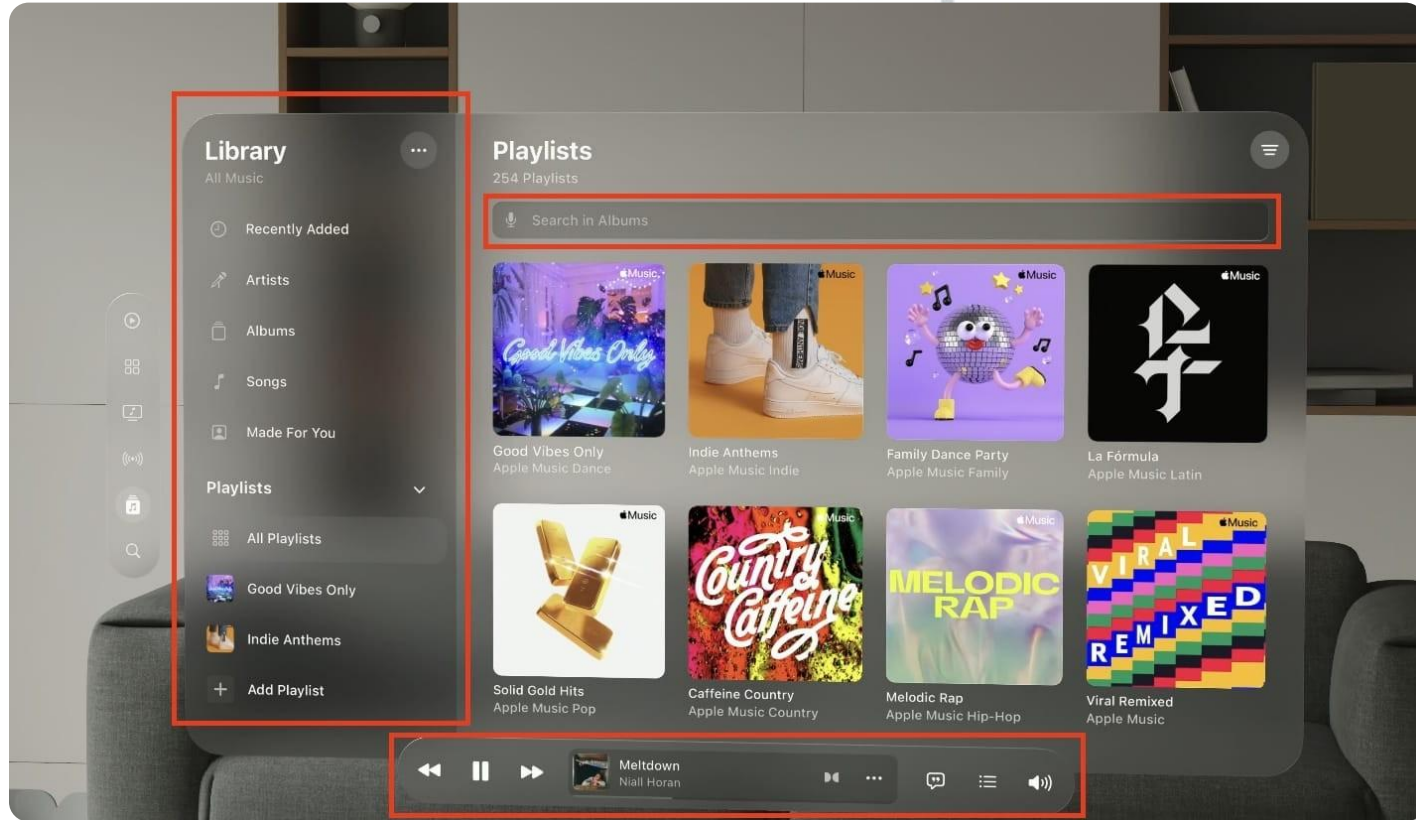
Checker Shadow Illusion



Edward H. Adelson

Our brain interprets lighting and shadows, we perceive them as different colours.

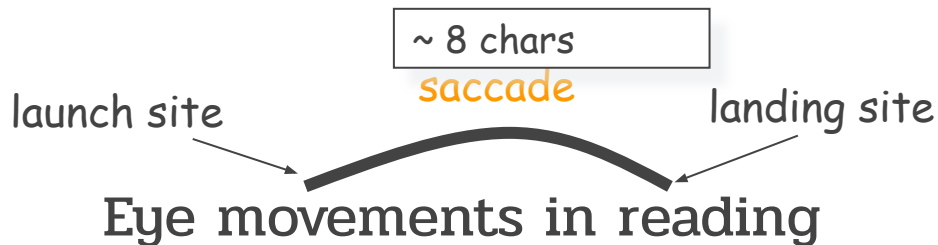
3D UI Design



How do we see the world and how do we read?



Eye movement in reading

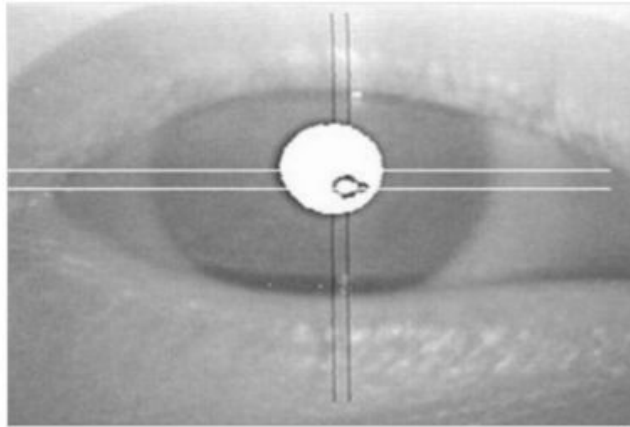


ชิมไป
หน้าจอนี้ยาวพอที่จะอ่านได้โดยไม่ต้องอ่านทีเดียว ๆ รังเกตุเห็นนางกัม
สหายตาของนางแยลงทอวัน จนกระทั่งแนวสายตาซึ่งตกอยู่ที่เส้นเดิมก็
ชาวยอะไรไม่ได้มอง หน้าข้างบางครั้งยังกลายเป็นสิ่งกะกะกิดก่อให้เกิด
ความรำคาญมิได้มีอหังสองข้างที่ยังคงทำงานของมันด้วยความกระตือรือร้น
แต่ถึงกระนั้นก็ยังยวบยออยู่ไม่อยู่ในเวลาที่ควรต้องกลิ้งเปลี่ยนด้านใน
เพื่อจะไปให้คนเขียนเป็นเนื้อเดียวกัน นางรู้ตัวว่าหน้าเดียวกันนางประหลาดนี้
ได้กลายเป็นสิ่งที่ยากเย็นและทำได้ไม่ประณีตเหมือนเมื่อก่อน บ่อยครั้ง
ที่นางปฏิเสธที่จะรับชมหน้าทั้ง ๆ ที่ได้เจตนาที่จะอ่านไปปะ หรือสอยเลื้อ

Eye tracking

The eye tracking monitor uses infrared light to **make the pupils of the person sitting in front of it light up** and so become detectable.

This is what it looks like for the monitor.



tobi

OptimalUsability



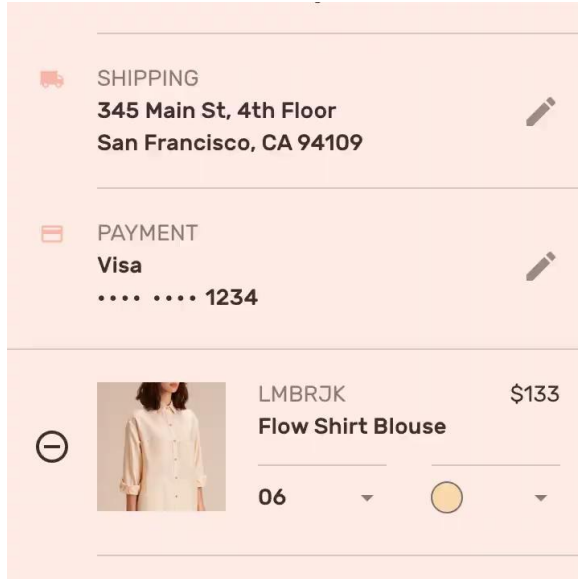
How we look on screen



Why Sound Matters in UI Design?



Sound in UI design



Notifications and alerts



Navigation



Emotion

McGurk Effect



Touch

- A touch screen is a computer display screen that is sensitive to human touch, allowing a user to interact with the computer by touching pictures or words on the screen.
- At present there are many digital devices use sense of touch to control the interface.



How we hold our smartphone



49%

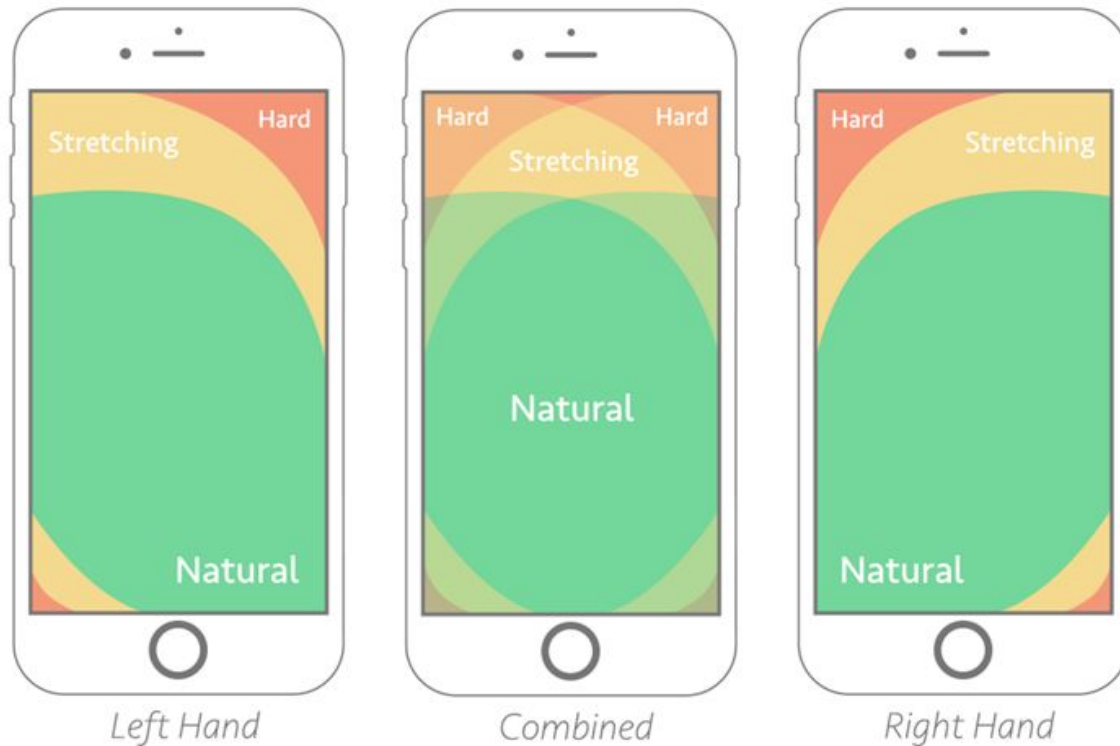


36%

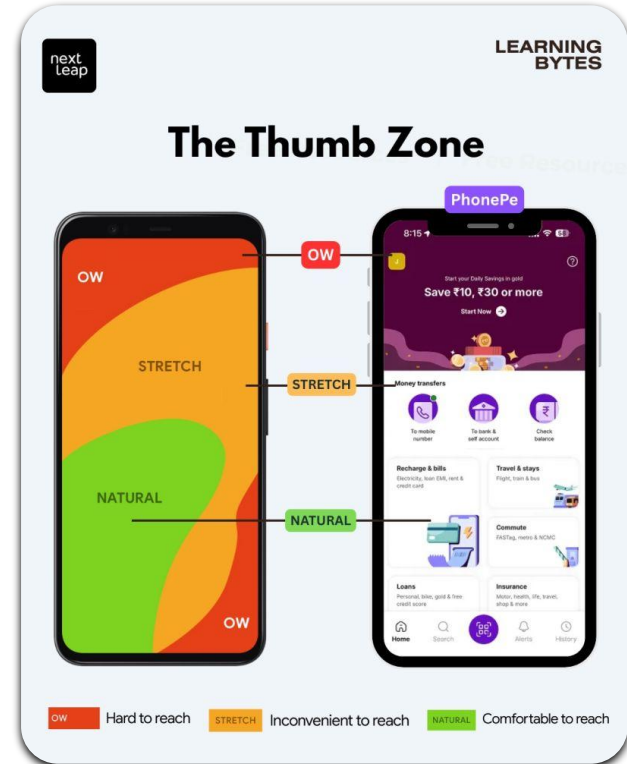
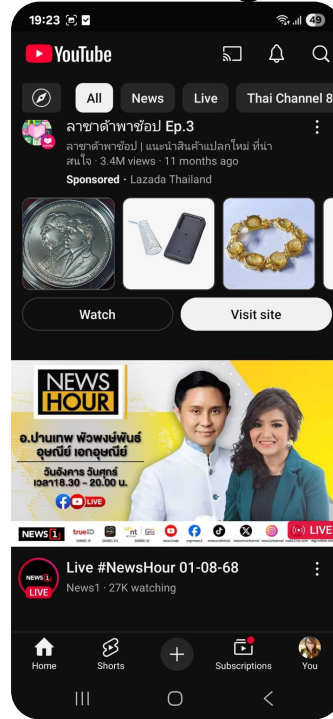
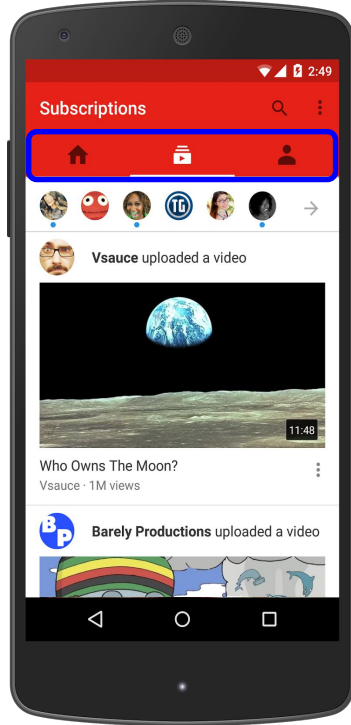


15%

The Thumb Zone: Designing For Mobile Users



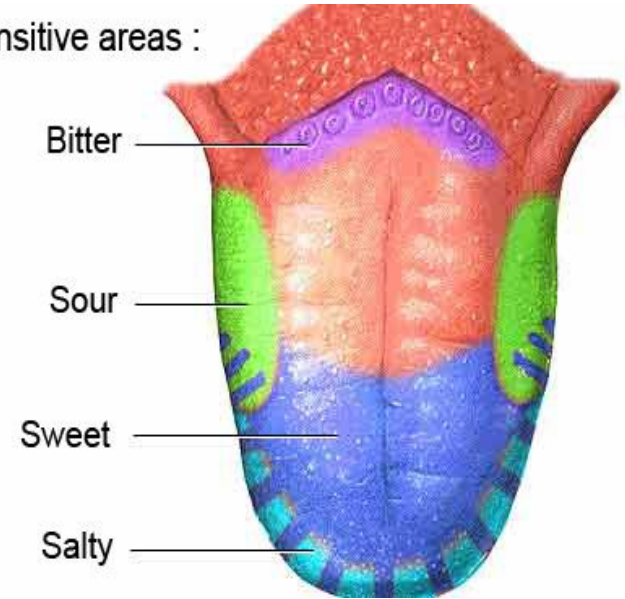
The Thumb Zone: Designing For Mobile Users



Taste



Taste sensitive areas :



Taste



Smell - Olfactory interfaces



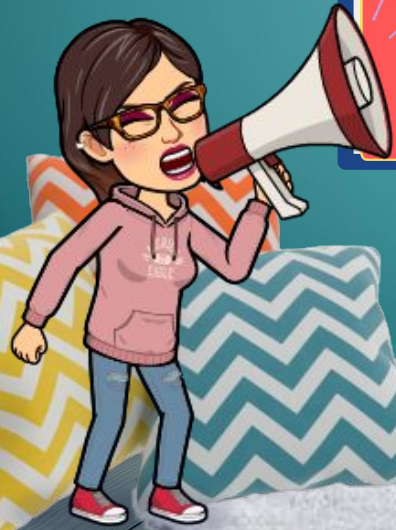
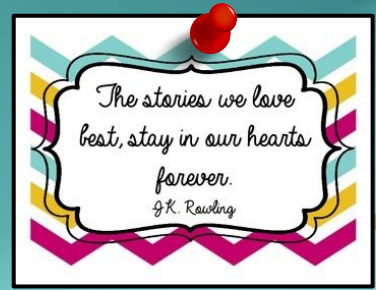


Checking your attendance

Week 2 Lecture 2



Note that you will get 1% for answering the question correctly.



Contact me

You can contact me via :



Inthraporn.aranyanak@tudublin.ie

